

PATRICK ATTEY-YEBOAH

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EDUCATION

MPhil Epidemiology (Environmental)

2024-Present

University of Cape Coast, Ghana

Thesis: "Spatial variability in the association of ambient PM_{2.5} exposure with child undernutrition in sub-Saharan Africa"

BSc Biomedical Sciences

2017-2021

University of Cape Coast, Ghana

RESEARCH INTEREST

My research investigates the health impacts of ambient air pollution across Sub-Saharan Africa. I specialize in integrating environmental exposure data with large-scale population health datasets to generate policy-relevant evidence. My work combines spatial epidemiology, Bayesian statistical modeling, and data science approaches to examine outcomes such as child undernutrition and respiratory health, with a broader aim of advancing environmental health research in low-resource settings and translating findings into actionable policy insights.

EXPERIENCE

Graduate Research Assistant,

2023-Present

Public Health Research Group,

University of Cape Coast, Ghana

- Design and implement environmental epidemiology studies integrating air pollution exposure data with population health datasets.
- Develop and apply exposure assessment protocols using satellite-derived and ground-based PM_{2.5} measurements.
- Conduct statistical and geospatial analyses to model and assess environmental health risks.
- Air pollution modeling to characterize spatial and temporal variability in urban and regional settings.
- Support air quality monitoring campaigns, including data cleaning, validation, and integration of multi-source datasets.
- Lead and contribute to manuscript preparation, including study design, data analysis, and scientific writing for peer-reviewed journals.
- Collaborate with multidisciplinary teams to translate research findings into policy-relevant insights for public health interventions.

Breathe Accra Project & Ghana Urban Air Quality (GhanaAQ), Ghana

- Manage and analyze large-scale air quality monitoring datasets from low-cost sensor networks and reference instruments.
- Identify spatial and temporal trends, pollution hotspots, and exposure patterns using geospatial and statistical techniques.
- Develop and implement air quality models to improve exposure estimation in data-sparse environments.
- Transform complex datasets into policy-relevant outputs, including dashboards, reports, and decision-support tools.
- Ensure data quality assurance and control (QA/QC) through systematic validation and harmonization of multi-source data.
- Contribute to scientific manuscripts, reports, and technical documentation for environmental health research.
- Collaborate with researchers, policymakers, and stakeholders to support evidence-based air quality management strategies.

PUBLICATIONS

1. **Attey-Yeboah, P.***, Afful, C., Yeboah, K., Korkpoe, C. H., Coker, E. S., Subramanian, R., & Amegah, A. K. (2025). Utility of low-cost sensor measurement for predicting ambient PM 2.5 concentrations: evidence from a monitoring network in Accra, Ghana. *Environmental science: atmospheres*, 5(4), 517-529. <https://doi.org/10.1039/D4EA00140K>
2. **Attey-Yeboah, P.***, Adjorlolo, P. K., Aheto, J. M. K., Molitor, J., Coker, E. S., & Amegah, A. K. (2025). *Spatial variability in the association of ambient PM2.5 exposure with child undernutrition in sub-Saharan Africa*. *Environment International*, 109995. <https://doi.org/10.1016/j.envint.2025.109995>
3. Adjorlolo, P. K., **Attey-Yeboah, P.***, Molitor, J., Coker, E. S., Aheto, J. M., & Amegah, A. K. (2026). Spatial variability in the association of ambient PM2.5 exposure with acute respiratory infection among children in Sub-Saharan Africa. *Environmental Research*, 124297. <https://doi.org/10.1016/j.envres.2026.124297>
4. Amegah, Adeladza Kofia; Yeboah, Kelvin; Quarshie, Evans; Benjamin, Divine Onyekachia; Amonoo, Patricia; **Attey-Yeboah, Patrick**; Stranges, Saverio; Kandala, Ngianga-Bakwin. Consumption of antioxidant-rich foods attenuates respiratory and cardiovascular health effects in a population highly exposed to air pollution. *Environmental Epidemiology* 10(3):p e477, June 2026. | <https://doi.org/10.1097/EE9.0000000000000477>

UNDER PEER-REVIEW: **Attey-Yeboah, P.***, Afful, C., Yeboah, K., Coker, E. S., & Amegah, A. K. Utility of low-cost sensor data for air pollution source apportionment: Evidence from a major sub-Saharan African city.

CONFERENCES

- **India Clean Air Summit (ICAS) + CAMS-Net 2024**, (Poster Presentation) - "Utility of low-cost sensor measurement for predicting ambient PM 2.5 concentrations: evidence from a monitoring network in Accra, Ghana"

- **Joint International Society of Exposure Science and the International Society for Environmental Epidemiology Conference (ISES-ISEE) 2025**, (Poster Presentation) – “Spatial variability in the association of ambient PM_{2.5} exposure with child undernutrition in sub-Saharan Africa”.
- **Data for Health in Africa Meeting & 5th DS-I Africa Consortium Meeting (2025)**, (ECR Presentation) – “Spatial variability in the association of ambient PM_{2.5} exposure with child undernutrition in sub-Saharan Africa”

RESEARCH PROJECTS

Leveraging Data Science Applications to Improve Children's Environmental Health in Sub-Saharan Africa (DICE)

- **Funded by** National Institute Of Environmental Health Sciences of the National Institutes of Health under Award Number U01ES036147
- **Led by** Prof Kofi A Amegah
- **Role:** Research Student
- **Responsibilities:** Analysis and Manuscript Preparation for *Aims 1 (To establish the spatial variability in the impact of ambient PM_{2.5} exposure on children's health in SSA and explore the effect modifying role of neighbourhood greenness and nutrition.) and Aims 2 (To estimate ambient PM_{2.5} exposures at multi-temporal scales by integrating land use regression (LUR) models, high-resolution ground monitoring data, and mobile monitoring data in Uganda and Ghana)* of Project

TECHNICAL SKILLS

Programming Language: R Programming, Python

Others: Air Pollution Exposure Modelling, Data Analysis and Visualization, Machine Learning Modelling

REFERENCE

Kofi Adedza Amegah, PhD

Associate Professor of Environmental & Nutritional Epidemiology

Head of Biomedical Sciences Department

Vice-Dean of School of Public Health

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